On Normative Uncertainty

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Abstract

In this essay I study the problem on normative uncertainty. The focus is two-fold: on moral uncertainty, pertaining to which moral norms to follow; on decision-theoretic uncertainty, pertaining to which decision theory is correct. I also give careful consideration to higher-order uncertainty, i.e. uncertainty about norms purporting to govern our choices under uncertainty, uncertainty about those norms too, and so on. Throughout, I approach these topics in a semi-formal manner by sketching a mathematical model of the objects and their relations. Short of giving a complete philosophical analysis, the modest intent is to provide a preliminary picture of (some of) these notions by means of systematic treatment.

1 Introduction

Knowing what's morally right is a hard question. Indeed, we might wonder which moral norms should govern – and which do govern – our decisions. Must we worry about the intent behind our acts, or merely about their consequences? Need we be virtuous persons? Is there a morally relevant distinction between killing and letting die? And, if there were such a distinction, ought we take it into account when deliberating about our choices? Should we care about the well-being of every possible person? Or care only about every person that will necessarily exist? Should we hold an actualist view instead? The questions go on and on.

So far we have mentioned uncertainty of which moral norms to follow, or, alternatively, which utility function we should maximise expected utility with respect to. Uncertainty, however, can exist in different forms. We can be uncertain about how we should maximise expected utility – causally, evidentially, or by following some other procedure? – and even about whether we should be maximising expected utility in the first place. The question of utmost importance, here, is how we should make decisions when we're uncertain about how to make decisions.

In this essay we look at the problem of normative uncertainty from the point of view of moral uncertainty and decision-theoretic uncertainty. The focus, additionally, is on higher-order uncertainty. Just as we are uncertain about norms governing our choices, we may also be uncertain about norms governing our choices when we're uncertain about norms governing our choices, or even uncertain about norms governing our choices when we're uncertain about norms governing our choices when we're uncertain about norms governing our choices. And so on, and so forth.

2 Moral Uncertainty

I start by formalising a framework sketched in Macaskill [3, 5]; terms may differ superficially, and the notation is my own, but the underlying concepts are the same. This work is centred around the problem of moral

uncertainty, and builds on the earlier contributions of Ross [10], Lockhart [2], Sepielli [11], and several others.

In the next section, my efforts are towards formalising higher-order moral uncertainty, a concept discussed in Sepielli [12]. I don't adopt Sepielli's terminology, nor do I follow his philosophical musings, though I am working on problems closely related to his work, and besides that paper there's little of pertinence in the literature.

Thereafter, I illustrate the content previously introduced with examples, and extend the ideas a little further.

The focus of this essay is on formalisation, so the attentive reader is invited to consult Macaskill [3] (including the works cited therein) for a broader coverage of both moral uncertainty and inter-theoretic comparisons of value. The same can be said of the assumptions I adopt: every effort has been made to lay them out clearly, but the presentation is too concise for the material to be self-contained. Once again, the reader might benefit from further readings, e.g. Portmore [9] on consequentialising moral theories.

2.1 Assumptions

Throughout this essay, I will rely on three substantial assumptions: (i) the normative requirements of any moral theory can be expressed by an ordering over options; (ii) problems of moral decision-making can be stated so that the options available to the decision-maker are identical for all theories; (iii) inter-theoretic comparisons of value are sometimes possible across moral theories.

2.1.1 Consequentialising Moral Theories

The first assumption is related to the claim that moral theories (even Kantianism) can always be "consequentialised," viz. accounted for in the typical consequentialist framework that employs some notion of value to form an ordering over acts, outcomes, lotteries, or in our case, options. My statement is more specific: I assert that a mathematical structure,

namely an ordering over options, can be used to faithfully capture the normative requirements of any moral theory. I will postpone discussing the technical details, except to make it clear that this claim is a weak one. Though I'm sympathetic to several points in Peterson [8], I don't agree that an ordering must assign cardinal utilities to options; ordinal utility suffices here. And I allow for options to be incomparable in some cases, which amounts to rejecting completeness in the ordering they belong to. Furthermore, orderings can even be context-sensitive, so a moral theory may order some options differently depending on which other ones are available.

2.1.2 Common Options

The second assumption addresses the worry that different moral theories care about different aspects of an agent's choices, so they might not agree on which set of options is available to the decision-maker. Were it the case that problems of moral decision-making can be stated so that the set of available options is the same for all theories, then this worry would placated. I find such a claim to be reasonable, and I sketch a way of constructing sets of common options in a typical circumstance.

Take, for example, a decision problem where the uncertainty lies in the correctness of consequentialist or Kantian ethics. Consequentialism only cares about the consequences of an agent's choices, so it distinguishes options according to the state of affairs they bring about. Kantianism, in addition, explicitly takes into account the decision-maker's intent; for every option formulated in the consequentialist account, it may distinguish between different sub-options classified according the decision-maker's intent. So, to construct a common set of options, for each option formulated in the consequentialist account we take every combination of option/intent that is compatible with the original option. If a consequentialist theory cannot distinguish between the more fine-grained options, that's no problem: different option/intent pairs with the same original option will be treated as equally choice-worthy by that theory. Other cases will be harder than the example just discussed – they may involve far more com-

plex combinations of moral properties – but, regardless, the general idea is to specify options to the degree that every property taken into account by any moral theory is described in each option.

2.1.3 Inter-Theoretic Comparisons of Value

As for our last assumption, see MacAskill [3] for a thorough defence of the claim that comparisons of value are sometimes possible across moral theories; here we will merely illustrate that this assertion is reasonable, with a couple examples from that paper. Also, it goes without saying, whenever an inter-theoretic comparison of value is made in this essay the tacit assumption is that it falls under those cases where such comparisons are allowed.

First, let's look at a case where an inter-theoretic comparison of value is impossible – to establish that our strongest claim can only be existential, not universal. Consider a moral theory T that orders options based on a plurality of irreducible values, e.g. knowledge and beauty. Furthermore, suppose the theory explicitly states that these values are incomparable, in the sense that it's never meaningful to speak of a difference in knowledge as comparable to a difference in beauty. In such a theory options will, sometimes, be intra-theoretically incomparable, e.g. $o \equiv$ "knowledge" VS. $o' \equiv$ "beauty." Now, suppose a full inter-theoretic comparison were possible with another theory T' where options are always comparable intra-theoretically. If that were the case, then the value of o and o' under T could be compared, one by one, with that of the same options under T. But we also have that o and o' are comparable between themselves according to T', from which follows that o and o' must also be comparable under T, contradicting our initial assumption. We reach this conclusion by transitivity, after comparing o and o' under T, respectively, to o and o' under T' (by inter-theoretic comparability), and o to o' under T' (by intra-theoretic comparability). This counter-example is sufficient to establish that comparisons of value aren't always possible across moral theories.

As an aside, note that if we wish our models to stay faithful to theories that postulate intra-theoretic incomparability of options, like one just

described, then we must allow normative orderings to be incomplete – as mentioned previously.

A case that is favourable to inter-theoretic comparisons, instead, might involve belief revision on moral theories. Say you start believing in a theory T according to which killing is much worse than letting die: perhaps the difference isn't sharp or exactly known to you, but it is significant to speak of a difference in value between killing and letting die (according to T). Suppose you came to believe (with full certainty) that there's no morally relevant distinction between killing and letting die, while your other beliefs remain unchanged, then your credence could be split between: T' for which killing and letting die have the same value as killing according to T; T'' for which killing and letting die have the same value as letting die according to T. This is a situation in which you can clearly make comparisons of value across T, T' and T'', so inter-theoretic comparison of value are sometimes possible.

2.2 Moral Decision Problems

In this first instance, we will construct a formalism that captures decision-making under moral certainty. By decision-making under moral certainty I mean a situation described as follows: an agent, known as the decision-maker, has a set of options available to them; they believe in a moral theory that orders those options according to some normative constraints; as they believe in this moral theory with fully certainty, then they will choose options accordingly.

The problem we are about to model is simple, but the formal structures we introduce are at the core of all the models of decision-making we will study in this essay.

2.2.1 Options and Feasible Sets

The set O is the set of all options. Note that O isn't the set of options available to an agent in a specific decision situation, instead, it contains every option of every decision situation. A set of options available to an

agent in a specific decision situation, then, must be a subset of O; we call this set a feasible set. In formal notation, F is a feasible set for all $F \subseteq O$. So, $\mathcal{P}(O)$ is the set of feasible sets; it contains every set of options an agent may come across in a decision situation.

2.2.2 Relations and Orderings

An ordering of options is a binary relation R over a feasible set $F \subseteq O$. A binary relation, by definition, is a subset of the Cartesian product of the set over which it's defined, so we have $R \subseteq F^2 \subseteq O^2$. The set of all orderings over feasible sets, then, is just $\mathcal{P}(O^2)$. The interpretation we give to the relation is one of "choice-worthiness" (see MacAskill [3]), where options are ordered ascendingly according to how strong moral reason one has to choose them, relative to the criteria of some moral theory. So, an ordering captures the practical advice that a moral theory would give about choosing options from some feasible set. We treat this relation as a weak preference, though it can be alternatively defined as a strict one.

Choice-worthiness relations may or may not satisfy certain properties, e.g. completeness, according to which all options are comparable:

1. $\forall x, y \in O : R(x, y) \vee R(y, x)$

I won't even require that these relations be preorders, i.e. binary relations satisfying reflexivity (2.) and transitivity (3.):

2. $\forall x \in O : R(x, x)$
3. $\forall x, y, z \in O : R(x, y) \wedge R(y, z) \Rightarrow R(x, z)$

Transitivity of choice-worthiness is a substantial assumption, which I don't wish to make. Reflexivity, on the other hand, appears innocuous, so I believe we could grant it to all choice-worthiness relations. Still, there's no reason to commit to an unnecessarily strong position: for generality let's avoid fixing any constraints on the relation of choice-worthiness; they can always be added later, in accordance to the theories we model.

2.2.3 Moral Theories

Moral theories are treated as functions from the set of feasible sets $\mathcal{P}(O)$ to the set of orderings $\mathcal{P}(O^2)$ over feasible sets. More specifically, they are treated as functions that take a feasible set $F \subseteq O$ as an argument and return an ordering over all options $o \in F$, determined by some relation R. In formal notation, a moral theory is defined as $T_0 : \mathcal{P}(O) \nrightarrow \mathcal{P}(O^2)$ such that $T_0(F) \in \mathcal{P}(F^2)$ (or, equivalently, $T_0(F) \subseteq F^2$).

Our definition of moral theory might come across as narrow, but, as anticipated, the aim is merely to capture normative aspects – "moral theory" is somewhat of a term of art here. Obviously, an ordering will be constrained by whichever moral entities its theory postulates, e.g. permissions, obligations, virtue, agency, or intent. Beyond that, however, these entities are of no use to us, because the considerations they bring are irrelevant to normativity once we already have an ordering over options. Indeed, if theories were to give rise to the same normative requirements on the basis of different moral entities, then we would map these theories to identical orderings, while assigning separate credences to the theories themselves though.[1]

Lastly, note that moral theories were defined as partial functions, so they can be specified over a restricted domain. This is a sensible definition, as a moral theory may not express choice-worthiness judgements for all decision situations, and therefore, over all feasible sets.

2.3 Moral Uncertainty

Here we extend the previous formalism to cover decision-making where an agent is uncertain of which moral theory to follow. To model this new problem we will need to account for the decision-maker's beliefs over moral theories, as well as the ways they can choose a moral theory under this condition of uncertainty.

[1]The last point is more than a subtlety: take two moral theories (in the non-formal sense) that yield the same deontic verdict in every decision situation, one of which appeals to well-accepted notions, while the other argues that you should make certain choices by an unconvincing argument; these theories are associated with identical orderings, but it would be a mistake to deem them equally credible.

2.3.1 Credences

We can model the decision-maker's partial beliefs about moral theories, quite simply, through a credence function. This function is probability mass function mapping an event space Ω to the unit interval $[0, 1]$. For the event space we will take the set of all moral theories, defined formally as $\mathcal{H}_0 = \{T_i | i = 0\}$. So, the decision-maker's credence function is $p_0 : \mathcal{H}_0 \to [0, 1]$.

Just like any other probability mass function, it must meet the following requirements:

1. $p_0(\mathcal{H}_0) = 1$
2. $p_0(T_0 \cup T_0') = p_0(T_0) + p_0(T_0')$ for all $T_0, T_0' \in \mathcal{H}_0$ s.t. $T_0 \cap T_0' = \varnothing$

The latter condition, know as finite additivity, can be generalised to countable additivity:

2'. $p_0(\bigcup_{T_0 \in H} T_0) = \sum_{T_0 \in H} p_0(T_0)$ where $\forall T_0, T_0' \in H \subseteq \mathcal{H}_0 : T_0 \cap T_0' = \varnothing$

Of course, countable additivity isn't applicable to infinite sets of arbitrary cardinality; as the name suggests, it requires that $|\mathcal{H}_0| \leq |\mathbb{N}|$. Our theoretical framework is meant to cover infinite hypothesis classes, so countable additivity is normally assumed. In any given example, though, we will only look at a finite set of hypotheses, so finite additivity suffices for the purpose of illustration.

2.3.2 Choice-Procedures

We shall call choice-procedures the ways an agent can make decisions under moral uncertainty. From the decision-maker's credence function over moral theories, taken as the argument, a choice-procedure returns a moral theory. In formal notation we have $T_1 : P_0 \nrightarrow \mathcal{H}_0$, where $P_0 = \{p_i | i = 0\}$ is the set of credence functions over moral theories. The interpretation we give to this moral theory's orderings, of various feasible sets, is one of choice-worthiness given the situation of uncertainty (according, of course, to the choice-procedure that produced that moral theory).

Just as moral theories were treated as partial functions, the same should be done with choice-procedures: sometimes they are applicable only to certain kinds of uncertainty (e.g. what to do under moderate risk) or refer to specific moral theories (e.g. what to do when you're uncertain about theories on the moral status of animals), so total functions are inappropriate for modelling them.

2.4 Higher-Order Uncertainty

Here we extend the previous formalism to cover decision-making where an agent is also uncertain of which choice-procedure to follow. To model this new problem we will need to account for the decision-maker's beliefs over choice-procedures, as well as the ways they can choose a choice-procedure under this condition of uncertainty.

2.4.1 Choice-Procedures for Choice-Procedures

Just as an agent might be uncertain about which moral theory they should act in accordance to, they might also be uncertain about which choice-procedure they should follow when they are under moral uncertainty.

As before, the decision-maker's partial belief are modelled by a credence function. This time, however, the objects of belief are choice-procedures, which makes the set of choice-procedures $\mathcal{H}_1 = \{T_i | i = 1\}$ the event space. So, the credence function is $p_1 : \mathcal{H}_1 \to [0, 1]$.

To resolve their uncertainty, an agents needs a choice-procedure for choice-procedures. We define this as function in analogy to the previously constructed T_1, so we have $T_2 : P_1 \nrightarrow \mathcal{H}_1$ where $P_1 = \{p_i | i = 1\}$ is the set of credence functions over choice-procedures.

2.4.2 Choice-Procedures for Choice-Procedures for ...

Just as an agent might be uncertain about which moral theory they should act in accordance to, or which choice-procedure they should follow when they are under moral uncertainty, they might also be uncertain about what choice-procedure to adopt when they are uncertain about choice-

procedures for choice-procedures. And why stop here? Decision-making under uncertainty needn't be restricted to moral theories or choice-procedures, or even choice-procedures for choice-procedures; it can be generalised far above that.

The notation I introduced so far hints at higher-order functions, which, again, can be defined in analogy to the previous ones. For $n > 0$, we have: $\mathcal{H}_n = \{T_i | i = n\}$, which is the set of nth-order choice-procedures; $p_n : \mathcal{H}_n \to [0, 1]$, which is the decision-maker's credence function over nth-order choice-procedures; $P_n = \{p_i | i = n\}$, which is the set of credence functions over nth-order choice-procedures; $T_n : P_{n-1} \nrightarrow \mathcal{H}_{n-1}$, which is an nth-order choice-procedure. Note that T_n functions are defined as partial, for the same reason as first-order choice-procedures.

Out of convenience I shall use use the term 'theory' to refer to either moral theories or choice-procedures (so long as the context makes it unambiguous), or to both of these objects.

2.5 An Example

Finally, it's time to see how all the theory can applied to model a decision problem: we look at an example inspired by the work on animal ethics of Singer [13], similar to those in MacAskill [3] and Ross [10].

2.5.1 Dietary Uncertainty

I find meat to be far more delicious than vegetables: it's tasty, juicy, and succulent. Eating meat increases my psychological well-being, more so than eating vegetables does, which I think might be grounds for considering a meat-based diet more choice-worthy than a vegetarian one. But I also believe, more confidently, that eating meat is morally wrong. My beliefs are split between the aforementioned theories, respectively $Meat$ and Veg, such that $p_0(Meat) = 0.4$ and $p_0(Veg) = 0.6$; they are settled on any further, relevant aspect.

When faced with the choice of a salad or a steak, respectively o_{salad} and o_{steak}, what should I do? My options are clear: $F = \{o_{steak}, o_{salad}\}$. The

theories' orderings over those options are, also, unambiguous: $o_{salad} <_{Meat} o_{steak}$ and $o_{steak} <_{Veg} o_{salad}$. Yet, I'm not certain which moral theory I should abide to, so I need to follow a choice-procedure in order to make this decision.

Suppose my parents taught me, and I came to believe with confidence, something along the lines of: "when you don't know what's morally right, follow your parents' example." Call this theory Edu_1, and let $Edu_1 \in \mathcal{H}_1$. My parents are guilt-free meat eaters, i.e. $Edu_1(p_0) = Meat$, so according to Edu_1 I should eat meat. I also have my own opinion, however, which says I should act in accordance with the most likely theory according to my beliefs. Call this other theory Max_1, and also let $Max_1 \in \mathcal{H}_1$. Following Max_1's advice, I should act in accordance to the moral theory I believe most likely. The resulting option turns out to be o_{salad}, as $p_0(Meat) = 0.4 < 0.6 = p_0(Veg)$, and therefore, $Max_1(p_0) = \text{argmax}\{p_0(T)|T \in \mathcal{H}_0\} = Veg$.

I am left in situation of uncertainty, because my first-order theories have given me discordant advice: according to Edu_1 I should eat meat, but according to Max_1 I should not. So, what should I do? Conveniently it turns out we only need to step up one level for a solution.

Upon close examination I realise my beliefs haven't been shaped much by my upbringing, viz. $p_1(Edu_1) = 0.2$ and $p_1(Max_1) = 0.8$. The advice of Max_1 holds for first-order uncertainty, and above that too; formally, $Max_i \in \mathcal{H}_i$ if, and only if, $i > 0$. My parents, on the other hand, never taught me how to make decisions under higher-order uncertainty; their advice was only meant for moral uncertainty, and therefore, is not applicable above that. This additional information gives us $Edu_i \notin \mathcal{H}_i$ for $i \neq 1$. The only theory left in my second-order hypothesis class $\mathcal{H}_2$ is Max_2, for which $p_2(Max_2) = 1$. Once again Max_2 advises me to choose o_{salad}, as $Max_2(p_1) = \text{argmax}\{p_1(T)|T \in \mathcal{H}_1\} = Max_1$, and therefore, $Max_2(p_1)(p_0) = Max_1(p_0) = Veg$. My dietary uncertainty has been resolved at last: I should avoid eating meat.[2]

[2]Note that this example is representative only because we have assumptions (i) and (ii) in place. The former allows us to always consequentialise moral theories and construct orderings for them, while the latter guarantees that any decision situation can be phrased with a set of

2.6 Expected Moral Utility

This section is only meant to give a quick overview of some ideas that extend the formalisms previously described; at no point will I attempt to address criticism of the expected utility approach to moral decision-making, or even give a rigorous account of utility and preferences. Throughout, we will revisit the dietary example to illustrate new theories, but also reject old ones with an argument along the lines of MacAskill [3].

2.6.1 Utility

My dietary dilemma was resolved to what I deem to be a satisfactory outcome, but there's a sense in which I got the "right" answer for the "wrong" reasons, so to speak. Namely, I didn't take into account the degree to which an option is morally choice-worthy: the belief that allowing a sentient being to be slaughtered for my own culinary pleasure is morally wrong, on a par with murder, should have a different strength than, e.g., the belief that eating meat is morally wrong because meat production is wasteful from an economic point of view (as it diverts resources that could be used so that choice-worthy states of affairs obtain). This disregard for the magnitude of choice-worthiness is a feature of the Max_1 and Max_2 choice-procedures, which, therefore, we no longer adopt.

We can model magnitudes of choice-worthiness with a utility function mapping each outcome to its respective utility. Formally, we have $u : O \nrightarrow \mathbb{R}$. The interpretation we give to utility is one of choice-worthiness – greater the utility, greater the choice-worthiness – so a moral theory $T \in \mathcal{H}_0$ that takes into account utilities will order options such that $\forall o, o' \in O : o \leq_T o'$ if, and only if, $u_T(o) \leq u_T(o')$. Note that the utility function is a partial function, as, in any given decision situation, it only needs to be defined over the feasible set.

Now that we have the right tools, we can reformulate the dietary example with my credences split evenly between the aforementioned theories. By Veg we denote the theory that derives the moral status of eating meat

commonly available options.

from economic criteria alone, without regard for animal suffering, while Veg^+ will stand for the moral theory that incorporates this additional consideration and, accordingly, equates eating meat with murder; my beliefs across the two theories agree on every other relevant aspect, including that murder is extremely un-choice-worthy. As Veg and Veg^+ are both moral theories we have $Veg, Veg^+ \in \mathcal{H}_0$, of course, just like $Meat \in \mathcal{H}_0$. Additionally, we stipulated that my credences were split evenly between these two theories, so $p_0(Veg) = 0.3 = p_0(Veg^+)$; as before we have $p_0(Meat) = 0.4$. Lastly, we must fix appropriate utilities to the outcomes in accordance to each theory:[3] $u_{Meat}(o_{salad}) = 0$, $u_{Meat}(o_{steak}) = 1$, $u_{Veg}(o_{salad}) = 0$, $u_{Veg}(o_{steak}) = -10$, $u_{Veg^+}(o_{salad}) = 0$, and $u_{Veg^+}(o_{steak}) = -1 \cdot 10^6$.

Note a further inadequacy of the Max_1 choice-procedure: with my credence in the choice-worthiness of vegetarianism split across multiple theories, Max_1 will no longer advise me to eat the salad. Specifically, $Max_1(p_0) \in \text{argmax}\{p_0(x)|x \in \mathcal{H}_0\} = \{Meat\}$, as $p_0(Meat) = 0.4 > 0.3 = p_0(Veg) = p_0(Veg^+)$. But that's absurd, as my confidence in the choice-worthiness of vegetarianism hasn't decreased just because I've specified a moral theory in more detail.

So, can we find a choice-procedure that takes into account the degree to which options are morally choice-worthy and avoids the other pitfall of Max_1? The natural candidate is moral hedging, which (as the name suggests) advises to hedge bets across moral theories. But for a definition we must extend our formalisms, since this isn't a choice-procedure.

2.6.2 Moral Decision Theories

We previously defined moral theories and choice-procedures; now we introduce an extension of these concepts. If we take a step back a moment, we can see a key difference between moral theories and choice-procedures. Namely, a moral theory prescribes an option to the decision-maker out of a set of available ones (by giving an ordering of choice-worthiness), while a choice-procedure resolves an uncertainty in a subset of the agent's be-

[3] Note that these numbers – measuring choice-worthiness on a common scale – make sense only thanks to our assumption (iii) that inter-theoretic comparisons are sometimes possible.

liefs without prescribing any options. A moral decision theory, instead, shares characteristics of both these objects: it prescribes an option out of a set of available ones to the decision-maker, but, in doing so, it also takes into consideration a subset of the agent's beliefs. The local beliefs taken into consideration by an nth-order decision theory are all those pertaining to theories up to and including that order, e.g. a 0th-order decision theory will prescribe options based on the decision-maker's beliefs about moral theories, a 1st-order decision theory will also take into consideration whichever choice-procedures the agents gives credence to, and so on for higher-order theories.

Moral decision theories of order n are a function of theories up to the nth-order, which we defined formally as the set $\mathcal{H}_{\leq n} = \bigcup_{i=0}^{n} \mathcal{H}_i$. An agent's beliefs over such domain, then, can be modelled by credence functions $p_{\leq n} = \mathcal{H}_{\leq n} \rightarrow [0, 1]$, which form the set $P_{\leq n} = \{p_{\leq i} | i = n\}$. The decision-maker's credence function over theories up to the nth-order must be subject to all the constraint given by the individual credence functions defined at every level up to n, so we have the constraint $\forall T \in \mathcal{H}_i \subseteq \mathcal{H}_{\leq n} : p_{\leq n}(T) = p_i(T)$, where $0 \leq i \leq n$. Finally, we define an nth-order moral decision theory as a function that takes as arguments the feasible set and the agent's beliefs, and returns the set of maximally choice-worthy options out of those available. In formal notation, we have $T_{\leq n} : \mathcal{P}(O) \times P_{\leq n} \nrightarrow \mathcal{P}(O)$ such that $T_{\leq n}(F, p_{\leq n}) \subseteq F$.

2.6.3 Moral Hedging

Moral hedging takes into consideration the average utility of an option across moral theories, weighted by the likelihood of each theory, and prescribes the option with the greatest such expected utility. This moral decision theory can now be defined formally as $Hedge_{\leq 0}(F, p_{\leq 0}) = \text{argmax}$ $\{EU_0(o) | o \in F\}$, where $\forall o \in O : EU_0(o) = \sum_{T \in \mathcal{H}_0} p_0(T) \cdot u_T(o)$.

Let's revisit the dietary example with the advice of $Hedge_{\leq 0}$, focusing on uncertainty at the moral level, i.e. not taking into consideration choice-procedures for now. To hedge my bets I should choose the option – out of eating a salad or a steak – that maximises expected moral utility, which

we calculate as follows:

$$\begin{aligned}
EU_0(o_{salad}) &= p_0(Meat) \cdot u_{Meat}(o_{salad}) + p_0(Veg) \cdot u_{Veg}(o_{salad}) \\
&\quad + p_0(Veg^+) \cdot u_{Veg^+}(o_{salad}) \\
&= 0.4 \cdot 0 + 0.3 \cdot 0 + 0.3 \cdot 0 \\
&= 0
\end{aligned}$$

$$\begin{aligned}
EU_0(o_{steak}) &= p_0(Meat) \cdot u_{Meat}(o_{steak}) + p_0(Veg) \cdot u_{Veg}(o_{steak}) \\
&\quad + p_0(Veg^+) \cdot u_{Veg^+}(o_{steak}) \\
&= 0.4 \cdot 1 + 0.3 \cdot (-10) + 0.3 \cdot (-1 \cdot 10^6) \\
&\approx -3 \cdot 10^5
\end{aligned}$$

So, we have $Hedge_{\leq 0}(F, p_{\leq 0}) = \text{argmax}\{EU_0(o) | o \in F\} = \{o_{salad}\}$, since $EU_0(o_{steak}) \approx -3 \cdot 10^5 < 0 = EU_0(o_{salad})$. In other words, moral hedging advises me to eat the salad.

To consider the dietary example in full, however, we still need to account for my beliefs about higher-order theories. In the original example, once we discard Max_1 and Max_2, I still have some credence in the 1st-order choice-procedure Edu_1. For the purpose of illustration, let's stipulate this credence to be as before – i.e. $p_1(Edu_1) = 0.2$ – and that my belief in hedging has replaced my previously held belief in Max_1 and Max_2. This is no problem, as we can generalised the theory to higher levels. For $n \in \mathbb{N}$, $Hedge_{\leq n}$ is defined as $Hedge_{\leq n}(F, p_{\leq n}) = \text{argmax}\{EU_n(o) | o \in F\}$, where $\forall o \in O : EU_n(o) = \sum_{T \in \mathcal{H}_n} p_n(T) \cdot u_T(o)$. But expected moral utility won't always be defined, because $u_T(\cdot)$ might not be defined for some $T \in \mathcal{H}_n$ (especially when $n \in \mathbb{N}^+$), so moral hedging isn't applicable to all decision problems. Conveniently, though, this condition is met in our example: despite being a choice-procedure, Edu_1 selects $Meat$ which does have a utility associated with each outcome. Calculating the expected utility of the options, then, we get:

$$
\begin{aligned}
EU_1(o_{salad}) &= p_1(Edu_1) \cdot u_{Edu_1}(o_{salad}) + p_1(Hedge_{\leq 0}) \cdot u_{Hedge_{\leq 0}}(o_{salad}) \\
&= 0.2 \cdot u_{Meat}(o_{salad}) + 0.8 \cdot EU_0(o_{salad}) \\
&= 0.2 \cdot 0 + 0.8 \cdot 0 \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
EU_1(o_{steak}) &= p_1(Edu_1) \cdot u_{Edu_1}(o_{steak}) + p_1(Hedge_{\leq 0}) \cdot u_{Hedge_{\leq 0}}(o_{steak}) \\
&= 0.2 \cdot u_{Meat}(o_{steak}) + 0.8 \cdot EU_0(o_{steak}) \\
&\approx 0.2 \cdot 1 + 0.8 \cdot (-3 \cdot 10^5) \\
&\approx -2.4 \cdot 10^5
\end{aligned}
$$

We have $Hedge_{\leq 1}(F, p_{\leq 1}) \in \text{argmax}\{EU_1(o) | o \in F\} = \{o_{salad}\}$, since $EU_1(o_{steak}) \approx -2.4 \cdot 10^5 < 0 = EU_1(o_{salad})$. So, taking into consideration my higher-order beliefs, moral hedging still advises me to opt for the salad.

2.6.4 Global Theories

Moral decision theories, as we have defined them, are limited in providing advice, in that they take into consideration beliefs only up to some order n. But couldn't a theory prescribe options to an agent by taking into consideration all their credences? Yes, of course, a decision theory may span the totality of an agent's beliefs; we call such theories global.

Taking the set of all theories $\mathcal{H} = \bigcup_{i=0}^{\infty} \mathcal{H}_i$ as the domain, the decision-maker's beliefs can be modelled by a global credence function $p = \mathcal{H} \to [0, 1]$ which, of course, must be consistent with the individual credence functions defined at every level. Let $P = \{p \mid p : \mathcal{H} \to [0, 1]\}$ be the set of global credence functions. From these objects we can easily define a global moral decision theory, as $T : \mathcal{P}(O) \times P \nrightarrow \mathcal{P}(O)$ such that $T(F, p) \subseteq F$.

3 Decision-Theoretic Uncertainty

In this section we provide a rough outline of how to extend decision-making under uncertainty so that it takes into account decision-theoretic uncertainty too. This extension can be formalised similarly to higher-order moral decision-making; having already covered that extensively, here we don't provide an overly detailed account.

3.1 Higher-Order Decision Theory

Let's start by describing decision situations. There are four elements that constitute a decision situation, which may be represented by a quadruple $\langle \mathcal{A}, \mathcal{K}, p, u \rangle$. The feasible set $\mathcal{A}$ is the set of acts available to the decision-maker in a given situation. The set of dependency hypotheses $\mathcal{K}$ models the causal relations between acts and the utility of the outcomes they bring about; each dependency hypothesis is treated as a counterfactual conditional, and there is one such conditional for each act in the feasible set. The credence function p, modelling the agent's partial beliefs, is a probability function defined over combinations of dependency hypotheses and acts. The utility function u maps outcomes – pairs of a dependency hypotheses and acts – to the real numbers, with the usual interpretation that a greater number measures the greater desirability of an outcome.

At an abstract level, we can think of a decision theory as a function that maps decision situations to choice-sets; the latter are sets of available acts deemed to be maximally choice-worthy by the theory. Each of these choice-sets, then, must be a subset of the feasible set belonging to the decision situation taken as input, so we have $T(\langle \mathcal{A}, \mathcal{K}, p, u \rangle) \subseteq \mathcal{A}$ for all decision theories T.

We distinguish decision theories according to the degree of decision-theoretic uncertainty they account for, by dividing them into hypothesis classes defined as before, i.e. $\forall n \in \mathbb{N}^+ : \mathcal{H}_n := \{T_i | i = n\}$. In $\mathcal{H}_0$ we place 0th-order decision theories, which are those theories that don't take into account uncertainty of the decision-theoretic kind; included here are casual and evidential decision theory as we normally think of them. In

higher-order classes – $\mathcal{H}_i$ for $i \geq 1$ – we place theories that treat lower-order decision theories just like acts and states, assigning them credence and utility, and take these additional objects into account when prescribing choices. We shall see how 1st-order decision theories work in the coming example.

3.2 An Example

We now apply our approach to decision-making – a hierarchy of decision theories that take into account uncertainty about lower decision theories – to Newcomb's problem. Such a famous example has received its lion's share of attention in the philosophical literature, yet the discussion has been focused almost entirely on which decision theory is correct. Here, instead, we analyse the rational choice of agent who is uncertain about which decision theory is correct – casual or evidential – and is faced with having to choose how many boxes to pick.

3.2.1 Newcomb's Problem

We need a precise formulation of Newcomb's problem as our results will depend on the exact pay-offs offered, rather than on the causal structure alone, so we choose a specific version. The problem, as originally published in Nozick [6], goes as follows:

"*Suppose a being in whose power to predict your choices you have enormous confidence. (One might tell a science-fiction story about a being from another planet, with an advanced technology and science, who you know to be friendly, etc.) You know that this being has often correctly predicted your choices in the past (and has never, so far as you know, made an incorrect prediction about your choices), and furthermore you know that this being has often correctly predicted the choices of other people, many of whom are similar to you, in the particular situation to be described below. One might tell a longer story, but all this leads you to believe that almost certainly this being's prediction about your choice in the situation to be discussed will be correct.*

There are two boxes, (Bl) and (B2). (B1) contains $1000. (B2) contains either $1000000 ($M), or nothing. What the content of (B2) depends upon will be described in a moment.

$$(B1)\ \{\$1000\} \qquad\qquad (B2)\ \left\{\begin{array}{c}\$M\\ or\\ \$0\end{array}\right\}$$

You have a choice between two actions:

(1) taking what is in both boxes

(2) taking only what is in the second box.

Furthermore, and you know this, the being knows that you know this, and so on:

(I) If the being predicts you will take what is in both boxes, he does not put the $M in the second box.

(II) If the being predicts you will take only what is in the second box, he does put the $M in the second box.

The situation is as follows. First the being makes its prediction. Then it puts the $M in the second box, or does not, depending upon what it has predicted. Then you make your choice. What do you do?"

3.2.2 Decision-Theoretic Problems

Letting one-boxing and two-boxing be denoted by B_1 and B_2 respectively, and X as the proposition that the second box contains a million dollars, we calculate the expected utility of the options to see which one is prescribed by each theory.

According to evidential decision theory the expected utilities are:

$$\begin{aligned} EEU_0(B_1) &= p(\neg X|B_1)\cdot u(B_1\wedge\neg X) + p(X|B_1)\cdot u(B_1\wedge X)\\ &= 0\cdot\$0 + 1\cdot\$1,000,000\\ &= \$1,000,000 \end{aligned}$$

$$
\begin{aligned}
EEU_0(B_2) &= p(\neg X|B_2) \cdot u(B_2 \wedge \neg X) + p(X|B_2) \cdot u(B_2 \wedge X) \\
&= 1 \cdot \$1,000 + 0 \cdot \$1,001,000 \\
&= \$1,000
\end{aligned}
$$

We have $EEU_0(B_2) = \$1,000 < \$1,000,000 = EEU_0(B_1)$, therefore, evidential decision theory advises to one-box.

According to causal decision theory, instead, the expected utilities are:[4]

$$
\begin{aligned}
CEU_0(B_1) &= p(B_1 \mathrel{\Box\!\!\rightarrow} \neg X) \cdot u(B_1 \wedge \neg X) + p(B_1 \mathrel{\Box\!\!\rightarrow} X) \cdot u(B_1 \wedge X) \\
&= 0.5 \cdot \$0 + 0.5 \cdot \$1,000,000 \\
&= \$500,000
\end{aligned}
$$

$$
\begin{aligned}
CEU_0(B_2) &= p(B_2 \mathrel{\Box\!\!\rightarrow} \neg X) \cdot u(B_2 \wedge \neg X) + p(B_2 \mathrel{\Box\!\!\rightarrow} X) \cdot u(B_2 \wedge X) \\
&= 0.5 \cdot \$1,000 + 0.5 \cdot \$1,001,000 \\
&= \$501,000
\end{aligned}
$$

We have $CEU_0(B_1) = \$500,000 < \$501,000 = CEU_0(B_2)$, therefore, causal decision theory prescribes two-boxing. So, the two theories disagree on what the decision-maker should do; this is the usual dilemma in Newcomb's problem.

3.2.3 Meta-Decision-Theoretic Solutions

We may overcome the discordant advice of the two decision theories by taking into account our decision-theoretic uncertainty, and applying casual and evidential utility maximisation once more.

Let's stipulate that, at every level of the decision-theoretic hierarchy, our credences are evenly split between casual and evidential decision the-

[4]Here, as elsewhere, I'm adopting Gibbard and Harper's formulation of causal decision theory, which is one of many – see Lewis [1] for alternatives and how they are related. This approach goes hand in hand with the deterministic assumption that, for each act, there is an outcome that would certainly result were the decision-maker to choose that act (and the agent is aware of this fact). Ours are simplifying assumptions, on which nothing of too importance hinges; no doubt we can give a more general formulation that deals with indeterminism, but it's outside the scope of this paper.

ory, i.e. $\mathcal{H}_n = \{CDT_n, EDT_n\}$ and $p(CDT_n) = 0.5 = p(EDT_n)$ for $n \geq 0$. And, for $\forall n \in \mathbb{N}^+$ and $\forall A \in \mathcal{A} = \{B_1, B_2\}$, we calculate expected utility as follows:[5]

$$\begin{aligned} CEU_n(A) &= p(A \mathrel{\Box\!\!\rightarrow} EDT_{n-1}) \cdot u_{EDT_n}(EDT_{n-1} \wedge A) \\ &\quad + p(A \mathrel{\Box\!\!\rightarrow} CDT_{n-1}) \cdot u_{EDT_n}(CDT_{n-1} \wedge A) \\ &= p(EDT_{n-1}) \cdot EEU_{n-1}(A) + p(CDT_{n-1}) \cdot CEU_{n-1}(A) \end{aligned}$$

$$\begin{aligned} EEU_n(A) &= p(EDT_{n-1}|A) \cdot u_{EDT_n}(EDT_{n-1} \wedge A) \\ &\quad + p(CDT_{n-1}|A) \cdot u_{EDT_n}(CDT_{n-1} \wedge A) \\ &= p(EDT_{n-1}|A) \cdot EEU_{n-1}(A) + p(CDT_{n-1}|A) \cdot CEU_{n-1}(A) \end{aligned}$$

First-order causal decision theory has no degrees of freedom, so we can immediately calculate the causal expected utility of each option. Letting $n = 1$ in the equation for $CEU_n(\cdot)$ above gives us:

$$\begin{aligned} CEU_1(B_1) &= p(EDT_0) \cdot EEU_0(B_1) + p(CDT_0) \cdot CEU_0(B_1) \\ &= 0.5 \cdot 1,000,000 + 0.5 \cdot 500,000 \\ &= \$750,000 \end{aligned}$$

$$\begin{aligned} CEU_1(B_2) &= p(EDT_0) \cdot EEU_0(B_2) + p(CDT_0) \cdot CEU_0(B_2) \\ &= 0.5 \cdot 1,000 + 0.5 \cdot 501,000 \\ &= \$251,000 \end{aligned}$$

We have $CEU_1(B_2) = \$251,000 < \$750,000 = CEU_1(B_1)$, therefore CDT_1 prescribes B_1. One-boxing has greater the first-order causal expected utility, despite having lower object-level causal expected utility, so first-order causal decision theory advises to one-box. Unless first-order evidential decision theory disagrees – i.e. it advises to two-box – which

[5] We shall discuss the appropriateness of these formulæ at a later stage. But note that, in the formula for causal expected value, $p(EDT_{n-1}) = p(A \mathrel{\Box\!\!\rightarrow} EDT_{n-1})$ and $p(CDT_{n-1}) = p(A \mathrel{\Box\!\!\rightarrow} CDT_{n-1})$ since the correctness of a decision theory is counterfactually independent of whichever acts the decision-maker chooses.

option to pick is a settled matter. At the object level evidential decision theory prescribes one-boxing, but, as we just saw, that's no indication of what advice the same theory will give at the meta level. Let's investigate, then, when the first-order theories disagree, i.e. under which conditions $EEU_1(B_1) < EEU_1(B_2)$ holds, fixing the values already calculated.

Expanding the terms as summations we get the following:

$$\begin{aligned} &p(EDT_0|B_1) \cdot EEU_0(B_1) + p(CDT_0|B_1) \cdot CEU_0(B_1) \\ &\quad < p(EDT_0|B_2) \cdot EEU_0(B_2) + p(CDT_0|B_2) \cdot CEU_0(B_2) \end{aligned}$$

Or, equivalently, with the numeric values substituted:

$$\begin{aligned} &\$1,000,000 \cdot p(EDT_0|B_1) + \$500,000 \cdot p(CDT_0|B_1) \\ &\quad < \$1,000 \cdot p(EDT_0|B_2) + \$501,000 \cdot p(CDT_0|B_2) \end{aligned}$$

From this equation we can get a necessary condition, by substituting $1 - p(EDT_0|B_1)$ and $1 - p(EDT_0|B_2)$ for $p(CDT_0|B_1)$ and $p(CDT_0|B_2)$ respectively:

$$\begin{aligned} &\$1,000,000 \cdot p(EDT_0|B_1) + \$500,000 \cdot [1 - p(EDT_0|B_1)] \\ &< \$1,000 \cdot p(EDT_0|B_2) + \$501,000 \cdot [1 - p(EDT_0|B_2)] \end{aligned}$$

Which simplifies to:

$$\$500,000 \cdot p(EDT_0|B_1) - \$500,000 \cdot p(EDT_0|B_2) < \$1,000$$

And, thus, to:

$$p(EDT_0|B_1) + p(EDT_0|B_2) < 0.002$$

The end result, practically, amounts to $p(EDT_0|B_1) \approx 0 \approx p(EDT_0|B_2)$. The necessary condition is that conditionalising on either B_1 or B_2 must drastically decrease my credence in EDT_0, so any option I choose must be treated as strong evidence against evidential decision theory.

From here, substituting $1 - p(CDT_0|B_1)$ and $1 - p(CDT_0|B_2)$ for $p(EDT_0|B_1)$ and $p(EDT_0|B_2)$ respectively, yields:

$$1 - p(CDT_0|B_1) + [1 - p(CDT_0|B_2)] < 0.002$$

Which reduces to:

$$1.998 < p(CDT_0|B_1) + p(CDT_0|B_2)$$

So $p(CDT_0|B_1) \approx 1 \approx p(CDT_0|B_2)$ holds as well, obviously. Since I had a uniform prior over the object-level decision theories – i.e. $p(CDT_0|B_1) = 0.5 = p(EDT_0|B_1)$ – then discovering my choice must make me redistribute the mass of my credence function so that it becomes concentrated on CDT_0. And, again, I would have to update my beliefs in such a manner regardless of which of the two options I chose.

We have found necessary conditions that are extremely demanding, therefore, in all but exceptional circumstances, we expect first-order evidential decision theory to prescribe one-boxing instead. None of the above is very surprising once we take into account stake-sensitivity: in this formulation of Newcomb's problem, as in many others, there's a $1,000 : 1$ ratio between some of the pay-offs. Indeed, were I highly confident in the correctness of evidential decision theory I might still one-box, due to the skewness of the rewards.[6] As our first-order decision theories agree, we may consider the problem of how many boxes to pick – not which object-level decision theory is correct – solved at the meta level. Of course, we could proceed to take into account uncertainty at the first-order level and so on, by further applying casual and evidential utility maximisation, but we already reached a solution at this level.

[6]Similar observations have been made before outside of academia, by Carl Shulman and Toby Ord – see their discussion on the forum Less Wrong: `http://lesswrong.com/r/discussion/lw/hpy/normative_uncertainty_in_newcombs_problem/`

3.3 Issues and Objections

Here we look at criticism of our approach to expected utility, which brings into question how we can account for decision-theoretic uncertainty when making decisions.

3.3.1 MacAskill's Equation

Our analysis of Newcomb's problem relied on the following definition of 1st-order evidential expected utility:

$$u_{EDT_1}(B_1) = p(EDT_0|B_1) \cdot u_{EDT_0}(B_1) + p(CDT_0|B_1) \cdot u_{CDT_0}(B_1)$$

The only published work that evaluates choices in Newcomb's problem based on a weighted average of 0th-order casual and evidential expected utility is a long-forgotten discussion in Nozick [7]. But this approach doesn't make any distinction of types: it conflates object and higher-order decision theories, by treating the above sum as the expected utility of a 0st-order decision theory. More recently, however, the same sum has been proposed as the expected utility of 1st-order evidential decision theory, in Macaskill [4].[7]

My understanding is that in MacAskill [4] the equation above is based (and could only be based) on the assumptions that follow. First, we have the definition of evidential expected utility applied over the partition $\{CDT_0, EDT_0\}$, which is:

$$\begin{aligned} u_{EDT_1}(B_1) =& p(EDT_0|B_1) \cdot u_{EDT_1}(EDT_0 \wedge B_1) \\ &+ p(CDT_0|B_1) \cdot u_{EDT_1}(CDT_0 \wedge B_1) \end{aligned}$$

Then, we also have:

$$u_{EDT_1}(EDT_0 \wedge B_1) = u_{EDT_0}(B_1)$$
$$u_{EDT_1}(CDT_0 \wedge B_1) = u_{CDT_0}(B_1)$$

[7] MacAskill never makes this assumption explicit, but reveals it when stating that his account is extensionally equivalent to Nozick's under certain conditions.

These equalities are instances of a more general equation, which I call MacAskill's equation (as it's the most substantial assumption used in his definition of u_{EDT_1}). According to this equation, for any factual proposition F:

$$\forall T_0 \in \mathcal{H}_0 : u_{EDT_1}(T_0 \wedge F) = u_{T_0}(F)$$

Or, more generally, one might even think that:

$$\forall n \in \mathbb{N}^+, T_n \in \mathcal{H}_n, T_{n-1} \in \mathcal{H}_{n-1} : u_{T_n}(T_{n-1} \wedge F) = u_{T_{n-1}}(F)$$

I did find myself convinced by these equations. After all, it's plausible that the value that 1st-order evidential decision theory assigns to some act, under the assumption that a 0st-order decision theory is correct, is just the value of that act according to the correct 0st-order decision theory. Alas, prima facie plausibility isn't enough, as we're about to see.

3.3.2 Briggs' Objection

A strong objection can be made to MacAskill's equation, credit for which goes entirely to Briggs. Starting from the definition of evidential expected utility (applied at a meta level), and the fact that X and $\neg X$ form a partition, we have:

$$u_{EDT_1}(A) = p(X|A) \cdot u_{EDT_1}(A \wedge X) + p(\neg X|A) \cdot u_{EDT_1}(A \wedge \neg X)$$

From here we may substitute $CDT_0 \wedge B_1$ for A, to get:

$$\begin{aligned} u_{EDT_1}(CDT_0 \wedge B_1) = {} & p(X|CDT_0 \wedge B_1) \cdot u_{EDT_1}(CDT_0 \wedge B_1 \wedge X) \\ & + p(\neg X|CDT_0 \wedge B_1) \cdot u_{EDT_1}(CDT_0 \wedge B_1 \wedge \neg X) \end{aligned}$$

Also, MacAskill's equation gives us:

$$u_{EDT_1}(CDT_0 \wedge B_1) = u_{CDT_0}(B_1)$$
$$u_{EDT_1}(CDT_0 \wedge B_1 \wedge X) = u_{CDT_0}(B_1 \wedge X)$$
$$u_{EDT_1}(CDT_0 \wedge B_1 \wedge \neg X) = u_{CDT_0}(B_1 \wedge \neg X)$$

So, by substituting these last terms on both the LHS and RHS, we have:

$$\begin{aligned} u_{CDT_0}(B_1) =& p(X|CDT_0 \wedge B_1) \cdot u_{CDT_0}(B_1 \wedge X) \\ &+ p(\neg X|CDT_0 \wedge B_1) \cdot u_{CDT_0}(B_1 \wedge \neg X) \end{aligned}$$

The proposition X could stand for anything at all, as it was arbitrary in our construction. "*Let it be the proposition that the closed box contains a million dollars. It seems reasonable to assume that conditional on B_1, the proposition that I one-box, CDT_0 (the proposition that CDT is the right 0th-order theory) is probabilistically independent of X (the proposition that the closed box contains a million dollars).*"[8] Using this independence assumption we get:

$$u_{CDT_0}(B_1) = p(X|B_1) \cdot u_{CDT_0}(B_1 \wedge X) + p(\neg X|B_1) \cdot u_{CDT_0}(B_1 \wedge \neg X)$$

But, with X being a dependency hypothesis, causal and evidential expected utility are defined as follows:[9]

$$u_{CDT_0}(B_1) = p(X) \cdot u_{CDT_0}(B_1 \wedge X) + p(\neg X) \cdot u_{CDT_0}(B_1 \wedge \neg X)$$

$$\begin{aligned} u_{EDT_0}(B_1) &= p(X|B_1) \cdot u_{EDT_0}(B_1 \wedge X) + p(\neg X|B_1) \cdot u_{EDT_0}(B_1 \wedge \neg X) \\ &= p(X|B_1) \cdot u_{CDT_0}(B_1 \wedge X) + p(\neg X|B_1) \cdot u_{CDT_0}(B_1 \wedge \neg X) \end{aligned}$$

So, unless the conditional and unconditional probabilities happen to

[8] Private correspondence with Rachael Briggs (October 10, 2013).

[9] Note that, despite disagreeing about the utility of acts, casual and evidential decision theory agree about the utility of outcomes, which is why we have $u_{EDT_0}(B_1 \wedge X) = u_{CDT_0}(B_1 \wedge X)$ and $u_{EDT_0}(B_1 \wedge \neg X) = u_{CDT_0}(B_1 \wedge \neg X)$ in the second equation.

coincide, we just derived that $u_{CDT_0}(B_1) = u_{EDT_0}(B_1) \neq u_{CDT_0}(B_1)$. MacAskill's equation, coupled with basic definitions from probability theory and decision theory, appears to be inconsistent; what we have is a disheartening result.

3.3.3 Remarks

If Briggs' objection is correct – as it seems to be – then our analysis of Newcomb's problem is flawed. More concerning, however, is the possibility that higher-order expected utility theory rests on a grave error, that our whole framework relies on some obscure term. Admittedly, even with $\exists n \in \mathbb{N}^+, T_n \in \mathcal{H}_n, T_{n-1} \in \mathcal{H}_{n-1}$ s.t. $u_{T_n}(T_{n-1} \wedge F) \neq u_{T_{n-1}}(F)$, the term $u_{T_n}(T_0 \wedge F)$ is mathematically well-defined, and so are $u_{EDT_n}(F) = \sum_{T \in \mathcal{H}_{n-1}} p(T|F) \cdot u_{EDT_n}(T \wedge F)$ and $u_{CDT_n}(F) = \sum_{T \in \mathcal{H}_{n-1}} p(F \mathrel{\Box\!\!\rightarrow} T) \cdot u_{CDT_n}(T \wedge F)$. But what non-mathematical entity does $u_{T_n}(T_{n-1} \wedge F)$ correspond to when it's not equivalent to $u_{T_0}(F)$? And how do we evaluate it, or can we even evaluate it at all? I have no satisfactory answers, yet.

4 Conclusion

This essay, I believe, has reached the goal it set out from the start, of providing a preliminary picture of normative uncertainty by means systematic treatment; we sketched formal accounts of both moral and decision-theoretic uncertainty, paying careful attention to higher-order norms, and even applied our theories to a few examples. The goal was modest, however, which means there are important details left deserving attention.

We also saw that higher-order expected utility theory is fraught with issues needing to be addressed. In particular we have that, for any proposition, its (unconditional) value given by some theory cannot be equated with its conditional value given by a meta theory conditionalised on that theory. Still, there's no need for pessimism: even in the worst-case scenario, that we cannot develop a technical apparatus for higher-order expected utility, it might be possible to ground meta decision theory on some criterion other than maximisation of expected utility.

Lastly, there are related matters worth investigating. We could try, for example, to construct a unified account of decision-making that combines uncertainty of both empirical and moral kinds (perhaps decision-theoretic uncertainty could be included too). I leave topics like this one for future work.

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